

Trigonometric inequalities

Solve the equation first, then read the arc between the two points.

LESSON 86–88

3 × 40 MIN

ALGEBRA 10, §28

EXTENSION BEYOND P1

LESSON CLOCK

15'

30'

30'

25'

30'

5

From points to arcs	15 min
$\sin x > a$	30 min
$\cos x > a$	30 min
tangent inequalities	25 min
Quadratic and compound	30 min
Homework	5 min

LEARNING OBJECTIVES

- Solve $\sin x > a$ and its three relatives on the unit circle.
- Write the answer as a family of intervals, not of points.
- Choose the arc correctly for a negative right-hand side.
- Solve simple compound and quadratic trigonometric inequalities.

TEXTBOOK

National	pp. 289–300
Cambridge	Pure Mathematics 1, pp. 117–120
Workbook	Exercise 5G

01

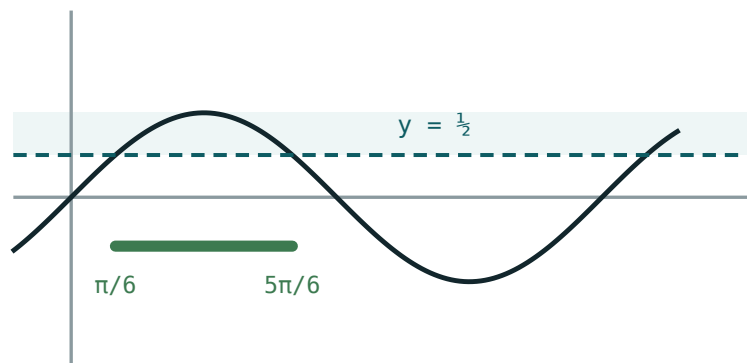
Explanation

From points to arcs

The equation $\sin x = a$ marks two points on the circle. The inequality $\sin x > a$ asks for the **arc** between them — the part of the circle lying above the chord.

THE METHOD, IN THREE LINES

- Solve the equation to find the two boundary points.
- Shade the arc on which the inequality is true.
- Read the arc from its start to its end, going **anticlockwise**, and add the period.



$$\sin x > \frac{1}{2} \Leftrightarrow \pi/6 + 2\pi k < x < 5\pi/6 + 2\pi k$$

$\sin x > \frac{1}{2}$ — the band above the line, and the intervals it cuts.

ALWAYS GO ANTICLOCKWISE

Read the arc in the direction of increasing angle. Written the other way, $\frac{5\pi}{6} < x < \frac{\pi}{6}$ is an empty interval, and the whole answer collapses.

$\sin x > a$ and $\sin x < a$

For $\sin x > a$, the arc runs from $\arcsin a$ anticlockwise to $\pi - \arcsin a$ — the upper arc.

$$\sin x > a \Leftrightarrow \arcsin a + 2\pi n < x < \pi - \arcsin a + 2\pi n$$

$$\sin x < a \Leftrightarrow \pi - \arcsin a + 2\pi n < x < 2\pi + \arcsin a + 2\pi n$$

INEQUALITY	BOUNDARY POINTS	ANSWER ON ONE PERIOD
$\sin x > \frac{1}{2}$	$\frac{\pi}{6}, \frac{5\pi}{6}$	$(\frac{\pi}{6}, \frac{5\pi}{6})$
$\sin x \geq \frac{\sqrt{2}}{2}$	$\frac{\pi}{4}, \frac{3\pi}{4}$	$[\frac{\pi}{4}, \frac{3\pi}{4}]$
$\sin x < \frac{1}{2}$	$\frac{\pi}{6}, \frac{5\pi}{6}$	$(\frac{5\pi}{6}, \frac{13\pi}{6})$
$\sin x > -\frac{1}{2}$	$-\frac{\pi}{6}, \frac{7\pi}{6}$	$(-\frac{\pi}{6}, \frac{7\pi}{6})$

Notice that the last line has the **longer** arc: a negative bound leaves most of the circle above the chord. If your interval for $\sin x > -\frac{1}{2}$ is shorter than half the circle,

you have shaded the wrong arc.

$\cos x > a$ and $\cos x < a$

Here the chord is **vertical**, and the boundary points are $\pm \arccos a$. The arc where the cosine is large lies on the right.

$$\cos x > a \Leftrightarrow -\arccos a + 2\pi n < x < \arccos a + 2\pi n$$

$$\cos x < a \Leftrightarrow \arccos a + 2\pi n < x < 2\pi - \arccos a + 2\pi n$$

INEQUALITY	ANSWER ON ONE PERIOD	LENGTH OF THE ARC
$\cos x > \frac{1}{2}$	$(-\frac{\pi}{3}, \frac{\pi}{3})$	$\frac{2\pi}{3}$
$\cos x < \frac{1}{2}$	$(\frac{\pi}{3}, \frac{5\pi}{3})$	$\frac{4\pi}{3}$
$\cos x \geq -\frac{\sqrt{2}}{2}$	$[-\frac{3\pi}{4}, \frac{3\pi}{4}]$	$\frac{3\pi}{2}$

A CHECK THAT COSTS NOTHING

The two arcs must add to 2π . If $\cos x > \frac{1}{2}$ gives $\frac{2\pi}{3}$ and $\cos x < \frac{1}{2}$ gives $\frac{4\pi}{3}$, the pair is consistent.

Tangent inequalities

The tangent has period π and a vertical asymptote in the middle of each period, so its solution intervals are **half** as long and never cross the asymptote.

$$\tan x > a \Leftrightarrow \arctan a + \pi n < x < \frac{\pi}{2} + \pi n$$

$$\tan x < a \Leftrightarrow -\frac{\pi}{2} + \pi n < x < \arctan a + \pi n$$

The interval is always **open** at $\frac{\pi}{2} + \pi n$, because the tangent is not defined there.

Writing $\leq \frac{\pi}{2}$ is a real error, not a small one.

THE PERIOD IS π , NOT 2π

Adding $2\pi n$ to a tangent inequality halves the solution set. Every tangent answer steps by π .

Quadratic and compound inequalities

Substitute exactly as for equations, solve the quadratic in t , then intersect with $-1 \leq t \leq 1$ and translate back.

Example. $2\sin^2x - \sin x - 1 > 0$. With $t = \sin x$: $(2t + 1)(t - 1) > 0$, so $t < -\frac{1}{2}$ or $t > 1$

1. The second is impossible, so the answer comes from $\sin x < -\frac{1}{2}$ alone.

INEQUALITY	IN t	SURVIVING CONDITION
$2\sin^2x - \sin x - 1 > 0$	$t < -\frac{1}{2}$ or $t > 1$	$\sin x < -\frac{1}{2}$
$\cos^2x \leq \frac{1}{4}$	$-\frac{1}{2} \leq t \leq \frac{1}{2}$	both branches
$4\sin^2x > 3$	$ t > \frac{\sqrt{3}}{2}$	two pairs of arcs

A compound inequality. $-\frac{1}{2} < \cos x < \frac{\sqrt{2}}{2}$ is an intersection of two arcs. Shade

both on one circle and read the overlap — two intervals, symmetric about the horizontal axis.

02 Worked examples

EXAMPLE 1 Solve $\sin x > \frac{\sqrt{2}}{2}$.

① $\sin x = \frac{\sqrt{2}}{2} \Rightarrow x = \frac{\pi}{4}, \frac{3\pi}{4}$

The boundary points.

② Shade the arc above the chord.

It runs anticlockwise from $\frac{\pi}{4}$.

③ $\frac{\pi}{4} + 2\pi n < x < \frac{3\pi}{4} + 2\pi n$

ANSWER

$$\frac{\pi}{4} + 2\pi n < x < \frac{3\pi}{4} + 2\pi n$$

EXAMPLE 2 Solve $\cos x \leq -\frac{1}{2}$.

① $\arccos(-\frac{1}{2}) = \frac{2\pi}{3}$

② The cosine is small on the **left** arc.

③ $\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{4\pi}{3} + 2\pi n$

Closed — the inequality is not strict.

ANSWER

$$\frac{2\pi}{3} + 2\pi n \leq x \leq \frac{4\pi}{3} + 2\pi n$$

EXAMPLE 3 Solve $\tan x \geq 1$.

$$\textcircled{1} \quad \arctan 1 = \frac{\pi}{4}$$

$$\textcircled{2} \quad \text{The tangent increases to } +\infty \text{ at } \frac{\pi}{2}.$$

$$\textcircled{3} \quad \frac{\pi}{4} + \pi n \leq x < \frac{\pi}{2} + \pi n$$

Open at the asymptote.

ANSWER

$$\frac{\pi}{4} + \pi n \leq x < \frac{\pi}{2} + \pi n$$

EXAMPLE 4 Solve $2\sin^2 x - \sin x - 1 > 0$ on $[0, 2\pi]$.

$$\textcircled{1} \quad (2t + 1)(t - 1) > 0$$

With $t = \sin x$.

$$\textcircled{2} \quad t < -\frac{1}{2} \text{ or } t > 1; \text{ the second is empty.}$$

$$\textcircled{3} \quad \sin x < -\frac{1}{2}$$

$$\textcircled{4} \quad \frac{7\pi}{6} < x < \frac{11\pi}{6}$$

ANSWER

$$\frac{7\pi}{6} < x < \frac{11\pi}{6}$$

03 Interactive model

Give each pair a paper circle and a ruler for the chord; they shade, then read the interval aloud.

From the sign of an expression to an interval

$x > 2$

A horizontal number line with tick marks from -8 to 8. An open circle is at 2, and the line is shaded to the right of 2.

inequality $x > 2$

boundary 2 (open)

$x = 0$ works? no

$x = 6$ works? yes

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O‘ZBEKCHA	РУССКИЙ
Inequality	Tengsizlik	Неравенство
Arc	Yoy	Дуга
Interval of solutions	Yechimlar oralig‘i	Интервал решений
Strict inequality	Qat‘iy tengsizlik	Строгое неравенство
Union of intervals	Oraliqlar birlashmasi	Объединение промежутков
Endpoint	Chetki nuqta	Концевая точка
Period	Davr	Период
Sign chart	Ishoralar jadvali	Таблица знаков

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

$\sin x > a$ is solved by reading:

two points

the arc between them

the whole circle

nothing

The arc must be read:

clockwise

anticlockwise

either way

from the larger point

$\cos x > \frac{1}{2}$ on one period is:

$(0, \frac{\pi}{3})$

$(-\frac{\pi}{3}, \frac{\pi}{3})$

$(\frac{\pi}{3}, \frac{5\pi}{3})$

$(0, 2\pi)$

A tangent inequality steps by:

2π

π

$\frac{\pi}{2}$

π or 2π

The interval for $\tan x \geq 1$ is:

closed at both ends

open at $\frac{\pi}{2}$

open at both ends

a single point

$\sin x > 2$ has:

one interval

two intervals

no solution

all of $\mathbb{R}$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Solve $\sin x > 0$

ANSWER $2\pi n < x < \pi + 2\pi n$

2. Solve $\cos x > 0$

ANSWER $-\frac{\pi}{2} + 2\pi n < x < \frac{\pi}{2} + 2\pi n$

3. Solve $\sin x < 0$

ANSWER $\pi + 2\pi n < x < 2\pi + 2\pi n$

4. Solve $\tan x > 0$

ANSWER $\pi n < x < \frac{\pi}{2} + \pi n$

5. Solve $\cos x \geq 1$

ANSWER $x = 2\pi n$

6. Has $\sin x \geq 1.2$ a solution?

ANSWER No

7. Solve $\sin x \leq 1$

ANSWER All x

Medium · the standard question

7 PROBLEMS

1. Solve $\sin x > \frac{1}{2}$

ANSWER $\frac{\pi}{6} + 2\pi n < x < \frac{5\pi}{6} + 2\pi n$

2. Solve $\cos x < \frac{1}{2}$

ANSWER $\frac{\pi}{3} + 2\pi n < x < \frac{5\pi}{3} + 2\pi n$

3. Solve $\sin x \geq \frac{\sqrt{3}}{2}$

ANSWER $\frac{\pi}{3} + 2\pi n \leq x \leq \frac{2\pi}{3} + 2\pi n$

4. Solve $\cos x \leq -\frac{\sqrt{2}}{2}$

ANSWER $\frac{3\pi}{4} + 2\pi n \leq x \leq \frac{5\pi}{4} + 2\pi n$

5. Solve $\tan x < \sqrt{3}$

ANSWER $-\frac{\pi}{2} + \pi n < x < \frac{\pi}{3} + \pi n$

6. Solve $\sin x > -\frac{1}{2}$

ANSWER $-\frac{\pi}{6} + 2\pi n < x < \frac{7\pi}{6} + 2\pi n$

7. Length of the arc for $\cos x > -\frac{1}{2}$

ANSWER $\frac{4\pi}{3}$

Hard · stretch and reason

7 PROBLEMS

1. Solve $2\sin^2 x - \sin x - 1 > 0$ on $[0, 2\pi]$

ANSWER $\frac{7\pi}{6} < x < \frac{11\pi}{6}$

2. Solve $4\cos^2 x \leq 3$ on $[0, \pi]$

ANSWER $\frac{\pi}{6} \leq x \leq \frac{5\pi}{6}$

3. Solve $\sin 2x > \frac{1}{2}$ on $[0, \pi]$

ANSWER $\frac{\pi}{12} < x < \frac{5\pi}{12}$

4. Solve $-\frac{1}{2} < \cos x < \frac{\sqrt{2}}{2}$ on $[0, 2\pi]$

ANSWER $(\frac{\pi}{4}, \frac{2\pi}{3}) \cup (\frac{4\pi}{3}, \frac{7\pi}{4})$

5. Solve $\sin x \cos x > 0$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{2}) \cup (\pi, \frac{3\pi}{2})$

6. Solve $\tan^2 x > 1$ on $(-\frac{\pi}{2}, \frac{\pi}{2})$

ANSWER $(-\frac{\pi}{2}, -\frac{\pi}{4}) \cup (\frac{\pi}{4}, \frac{\pi}{2})$

7. Solve $\sin x + \cos x > 1$ on $[0, 2\pi]$

ANSWER $(0, \frac{\pi}{2})$

07 Homework

Homework — 5 tasks

Draw and shade a circle for every question; the picture is half the mark.

1. Solve $\sin x \geq \frac{1}{2}$ and give the answer on $[0, 2\pi]$.
2. Solve $\cos x > -\frac{\sqrt{3}}{2}$, and state the length of the arc.
3. Solve $\tan x \leq -1$, taking care over the asymptote.
4. Solve $2\cos^2 x - \cos x - 1 < 0$ on $[0, 2\pi]$.
5. Explain, with one circle, why $\sin x > -0.9$ has a much longer solution arc than $\sin x > 0.9$.